(iii) Nuclear force does not act on the electrons. So it does not alter the chemical properties of the atom.

9.6 RADIOACTIVITY

In the binding energy curve, the stability of the nucleus that has $Z > 82$ starts to decrease and these nuclei are fairly unstable nuclei. Some of the unstable decay naturally by emitting certain particles to form a stable nucleus. The elements with atomic number $Z > 82$ and isotopes of lighter nuclei belong to the category of naturally-occurring radioactive nuclei. Each of these radioactive nuclei decays to another nucleus by the emission of ${}_2^4He$ nucleus ($\alpha-$decay) or electron or positron ($\beta-$decay) or gamma rays ($\gamma-$decay).

The phenomenon of spontaneous emission of highly penetrating radiations such as α, β and γ rays by an element is called radioactivity and the substances which emit these radiations are called radioactive elements. These radioactive elements can be heavy elements ($Z > 82$), isotopes of lighter and heavy elements and these isotopes are called radioisotopes. For example, carbon isotope ${}_6^{14}C$ is radioactive but ${}_6^{12}C$ is not.

Radioisotopes have a variety of applications such as carbon dating, cancer treatment, etc. When a radioactive nucleus undergoes decay, the mass of the system decreases – that is, the mass of the initial nucleus before decay is always greater than the sum of the masses of the final nucleus and the emitted particle. This difference in mass Δm, it appears as the energy according to Einstein's relation $E = |\Delta m| c^2$.

The phenomenon of radioactivity was first discovered by Henri Becquerel in 1896. Later, Marie Curie and her husband Pierre Curie did a series of experiments in detail to understand the phenomenon of radioactivity. In India, Saha Institute of Nuclear Physics (SINP), Kolkata is the premier institute pursuing active research in nuclear physics.

Note

During early days of nuclear physics research, the term 'radiation' was used to denote the emanations from radioactive nuclei. Now we know that α rays are in fact ${}_2^4He$ nuclei and β rays are electrons or positrons. Certainly, they are not electromagnetic radiation. The γ ray alone is electromagnetic radiation.

9.6.1 Alpha decay

When an unstable nucleus decay by emitting an α-particle (${}_2^4He$ nucleus), it loses two protons and two neutrons. As a result, its atomic number Z decreases by 2 and the mass number decreases by 4. We write the alpha decay process symbolically in the following way

$$ {}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4He \qquad (9.26) $$

Here X is called the parent nucleus and Y is called the daughter nucleus.

Example: Decay of Uranium ${}_{92}^{238}U$ to thorium ${}_{90}^{234}Th$ with the emission of ${}_2^4He$ nucleus (α-particle)

$$ {}_{92}^{238}U \rightarrow {}_{90}^{234}Th + {}_2^4He $$

As already mentioned, the total mass of the daughter nucleus and ${}_2^4He$ nucleus is always

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